UNIVERSITY OF CALIFORNIA,
IRVINE

A High and Variable Dimensional Measurement of the Z+jets Differential Cross Section with the ATLAS Experiment and Aritifical Intelligence

DISSERTATION

submitted in partial satisfaction of the requirements
for the degree of

DOCTOR OF PHILOSOPHY

in Physics and Astronomy

by

Kevin Greif

Dissertation Committee:
Daniel Whiteson, Chair
Andrew Lankford
Jonathan Feng

2026

DEDICATION

(Optional dedication page)

To Dario Amodei mostly I guess?

TABLE OF CONTENTS

	Page
LIST OF FIGURES	**v**
LIST OF TABLES	**vi**
ACKNOWLEDGMENTS	**vii**
VITA	**viii**
ABSTRACT OF THE DISSERTATION	**xi**
1 Introduction	**1**
1.1 Open Problems in Fundamental Physics	3
1.2 The Deep Learning Revolution	10
1.3 Thesis Roadmap	13
2 Quantum Chromodynamics at the Large Hadron Collider	**15**
2.1 Jets and Jet Substructure at the LHC	18
2.1.1 The QCD Lagrangian and Asymptotic Freedom	18
2.1.2 Parton Showers and Infrared and Collinear Divergences	20
2.1.3 Jets	21
2.2 Energy correlator observables	25
2.2.1 Definition and History	25
2.2.2 Energy Correlators as Jet Substructure Observables	25
2.2.3 Experimental Status	28
2.3 Intrinsic Dimensionality of Jets	30
2.3.1 The Manifold Hypothesis and Intrinsic Dimension	30
2.3.2 The Energy Mover's Distance	32
2.3.3 The Nearest-Neighbor Intrinsic Dimension Estimator	33
2.3.4 Generator-Level Results	35
3 Machine Learning and Artificial Intelligence	**38**
3.1 Overview: ML as a Paradigm Shift	38
3.2 Neural Networks and Deep Learning	41
3.3 Generative Models	41

3.4 Diffusion Models . . . 41
3.5 Density Ratio Estimation and Reweighting . . . 41
3.6 Uncertainty Quantification . . . 41

4 Jet Classification at the LHC **42**
4.1 Boosted Jet Tagging: Motivation and Landscape . . . 42
4.2 ATLAS Top Tagger: Training and Uncertainty Quantification . . . 42
4.3 The Future of Jet Classification . . . 42

5 Unfolding **43**
5.1 Unfolding as an Inverse Problem . . . 43
5.2 AI-Based Unfolding: Unbinned and High-Dimensional . . . 43
5.3 Variational Latent Diffusion Unfolding . . . 43
5.4 Omnifold . . . 43

6 Z+Jets Cross Section Measurement **44**
6.1 Measurement Overview . . . 44
6.2 Event Selection and Simulation Samples . . . 44
6.3 OmniFold Implementation . . . 44
6.4 Results: Four Physics Use Cases . . . 44
6.4.1 Use Case 1: [FILL IN Title] . . . 44
6.4.2 Use Case 2: [FILL IN Title] . . . 44
6.4.3 Use Case 3: [FILL IN Title] . . . 44
6.4.4 Use Case 4: [FILL IN Title] . . . 44
6.5 Discussion . . . 44

7 Conclusion **45**
7.1 Machine Learning is Driving LHC Physics Progress . . . 45
7.2 diHiggs: The Next Milestone . . . 45
7.3 Agentic AI and the Transformation of Physics Research . . . 45
7.4 Will AI Deliver the Next Particle Physics Discovery? . . . 45

Appendix A Supplementary Material **57**

LIST OF FIGURES

Page

1.1 A timeline of the theoretical prediction or explanation (green) and experimental discovery (red) of all Standard Model particles. **TODO: cite as fair use** 9

2.1 The elementary particles of the Standard Model. The dynamics of the 6 quarks and the 8 gluons (all gluon fields are represented by one box in the diagram), described by the theory of quantum chromodynamics, is most relevant to this thesis. 16
2.2 The gluon self-interaction vertices that arise from the non-Abelian structure of QCD. 18
2.3 A compilation of experimental measurements of the strong coupling α_s as a function of the energy scale Q, illustrating asymptotic freedom through the decrease of the coupling at high energies. **TODO: cite PDG** 19
2.4 The two-point energy correlator measured in CMS 2011 open data for high $p_T$jets, showing distinct free-hadron, transition, and quark/gluon regimes as a function of the angular scale R_L (called z in the text). Figure adapted from Ref. [65]. 26
2.5 Illustration of the scale dependence of the intrinsic dimension. At large length scales the data appear as a single point (zero dimensions); at smaller scales the geometry of the underlying data manifold becomes visible and the estimated dimension increases. **TODO: cite!** 31
2.6 Illustration of the NNID estimator in a simple two-dimensional space. For a reference point (center), the distances $r_{k,i}$ to the ith nearest neighbor and $r_{k,j}$ to the jth nearest neighbor define the ratio $\mu_{k,ij}$. In this particular case $i = 5$ and $j = 10$. The distribution of these ratios across many reference points encodes the intrinsic dimension of the data manifold. 34
2.7 Nearest-neighbor intrinsic dimension (NNID) as a function of the median EMD scale, computed at truth level for the MadGraph (purple) and Sherpa (gold) event generators used in Chapter 6. The three panels show the same curve over different ranges of the EMD axis to resolve fine structure. Differences between the generators are most pronounced at very low EMD scales. . 37

LIST OF TABLES

Page

2.1 Standard Model fields and their representations under the gauge group $SU(3)_c \times SU(2)_L \times U(1)_Y$. Representations are written as $(\mathbf{d_3}, \mathbf{d_2}, Y)$, where d_3 and d_2 are the dimensions of the $SU(3)_c$ and $SU(2)_L$ representations, and Y is the weak hypercharge under the convention $Q = T_3 + Y/2$. Three generations of fermions are implied; all carry identical quantum numbers. 17

2.2 Summary of published experimental measurements of energy-energy correlator observables. 29

ACKNOWLEDGMENTS

I would like to thank...

(You must acknowledge grants and other funding assistance.

You may also acknowledge the contributions of professors and friends.

You also need to acknowledge any publishers of your previous work who have given you permission to incorporate that work into your dissertation. See Section 3.2 of the UCI Thesis and Dissertation Manual.)

VITA

Kevin Greif

EDUCATION

Doctor of Philosophy in Physics and Astronomy **2026**
University of California, Irvine *Irvine, California*
Advisor: Daniel Whiteson

Bachelor of Science, Magna Cum Laude **2020**
University of Notre Dame *Notre Dame, Indiana*
Majors: Physics and Program of Liberal Studies

RESEARCH EXPERIENCE

Foundation models for particle physics **Ongoing**
Researched the application of pre-trained foundation models in particle physics. Compared their performance to equivariant neural networks, and investigated the proper composition of pre-training data to optimize scaling laws.

Artificial Intelligence based unfolding **Ongoing**
Developed novel unfolding methods based on generative models and contributed to community comparison studies. Studied the use of unbinned, highly differential cross section measurements for parameter estimation. Deployed Artificial Intelligence based unfolding to perform a hyper-differential cross section measurement of Z+jets production with the ATLAS detector.

Constituent based top tagging **2022–2024**
Benchmarked several constituent top-tagging algorithms on realistic detector simulation. Assessed systematic uncertainties in tagger performance. Prepared research-grade open data records for use by the broader particle physics community.

Calibration of the ATLAS New Small Wheel **2022–2023**
Calibrated the peak detector output (PDO) readout used in the ATLAS New Small Wheel muon detector. Participated in detector operations during Run 3.

Physics inspired neural networks **2018–2020**
Deployed a physically motivated neural network architecture on a classification problem in particle physics.

CMS track trigger development **2017–2019**

Improved the efficiency of cuts on track candidates in the CMS experiment's track trigger algorithm.

SELECTED PUBLICATIONS

Only includes papers in which I made a significant contribution, in reverse chronological order.

1. V. Breso-Pla et al.. "Explicit or Implicit? Encoding Physics at the Precision Frontier". *Preprint:* arXiv:2603.08802, 2026.

2. A. Petitjean et al.. "Generative unfolding of jets and their substructure". *Preprint:* arXiv:2510.19906, 2025.

3. A. Shmakov et al.. "Full event particle-level unfolding with variable-length latent variational diffusion". *SciPost Phys.* 18(4):117, 2025.

4. N. Huetsch et al.. "The landscape of unfolding with machine learning". *SciPost Phys.* 18(2):070, 2025.

5. ATLAS Collaboration. "Accuracy versus precision in boosted top tagging with the ATLAS detector". *JINST* 19(08):P08018, 2024.

6. A. Shmakov et al.. "End-to-end latent variational diffusion models for inverse problems in high energy physics". *NeurIPS Proceedings*, 2023.

7. K. Greif and K. Lannon. "Physics Inspired Deep Neural Networks for Top Quark Reconstruction". *EPJ Web Conf.* 245:6029, 2020.

PRESENTATIONS

Jul. 2025	"Forward folding versus unfolding in the age of ML", *ML4Jets 2025*
Nov. 2024	"Full Event Particle-Level Unfolding with Variable Length Variational Latent Diffusion", *ML4Jets 2024*
Nov. 2023	"Systematic Effects in Jet Tagging Performance for the ATLAS Detector", *ML4Jets 2023*
Nov. 2023	"End-to-End Latent Variational Diffusion Models for Unfolding LHC Events", *ML4Jets 2023*
Nov. 2022	"Constituent-Based Top-Quark Tagging with the ATLAS Detector", *ML4Jets 2022*
Nov. 2019	"Physics Inspired Deep Neural Networks for Top Quark Reconstruction", *CHEP 2019*

APPOINTMENTS AND MEMBERSHIPS

ATLAS Early Career Scientist Board **2024–Present**
Advocate for the interests of early career scientists within the ATLAS collaboration. Organize educational and career development events.

Phi Beta Kappa Society **2020–Present**
Sigma Pi Sigma Society **2020–Present**

AWARDS

Outstanding Physics Major **2020**
Physics Department, University of Notre Dame

Paul Chagnon Award in Physics **2020**
Physics Department, University of Notre Dame — recognizing service to the department and fellow students

Edward J. Cronin Award **2020**
Program of Liberal Studies, University of Notre Dame — awarded for the finest piece of written work submitted in an academic year

Hichwa Fellowship **2019**
Physics Department, University of Notre Dame

TEACHING EXPERIENCE

Teaching Assistant, University of California, Irvine **2020–2021**

- Physics 3A (Mechanics), Fall 2021
- Physics 3LB (Electricity and Magnetism Lab), Spring 2021
- Physics 7LC (Mechanics Lab), Winter 2021
- Physics 2 (Mathematical Methods in Physics), Fall 2020
- Computational Methods in Physics, Spring 2020

OUTREACH

ATLAS Virtual Visit Tour Guide **2022–2025**
Remote visits to the ATLAS experiment for students from around the world.

UCI Physics and Astronomy Blog **2020–2022**
Published accessible descriptions of new research from the UCI physics department.

ABSTRACT OF THE DISSERTATION

A High and Variable Dimensional Measurement of the Z+jets Differential Cross Section with the ATLAS Experiment and Aritifical Intelligence

By

Kevin Greif

Doctor of Philosophy in Physics and Astronomy

University of California, Irvine, 2026

Daniel Whiteson, Chair

Proton-proton collisions at the Large Hadron Collider (LHC) offer the opportunity to observe the interactions of fundamental particles at very high energy scales. The high collision rate and large number of final state particles produced per collision imply that the datasets produced by detectors such as the ATLAS experiment are large and highly dimensional. The complexity of these datasets calls for the use of novel data analysis techniques which can exploit all of the available information to illuminate known interactions and search for new ones. This thesis presents applications of artifical intelligence (AI) techniques to improve the physics results of the ATLAS experiment. It centers on a full phase measurement of the Z+jets production cross section at the LHC, where the cross section is measured differential in the kinematics of every final state charged particle through the use of an AI based unfolding algorithm. This is the first such measurement performed at the LHC, which provides a complete experimental characterisation of the Z+jets production process and a proof-of-principle for the further pursuit of such measurements on a host of LHC processes.

Chapter 1

Introduction

The discovery of the Higgs Boson, announced on July 4th, 2012 [1, 2], is widely regarded as the most important discovery in particle physics in the 21st century. With that discovery the electroweak symmetry breaking mechanism [3–5] received experimental confirmation and the Standard Model of particle physics [6, 7] was complete. This triumph of modern science was rightly front-page news, and is still well known in the popular imagination. A less well known, though equally important, advancement was announced 87 days later on September 30th, 2012 when the results of the "ImageNet Large Scale Visual Recognition Challenge" were released. This was a challenge within the computer vision research community to develop a machine learning model that could sort images into 1000 classes. The training set for this challenge consisted of 1.2 million painstakingly hand-labeled images, which was unprecedented scale for labeled datasets at the time. This scale and the recent release of Nvidia's CUDA toolkit, which allowed for optimized matrix operations on GPU hardware, allowed for a team of researchers from the University of Toronto to train a deep convolutional neural network named AlexNet that outperformed its closest competitor by 10% in top-5 error rate[1] [8]. Before this moment state-of-the-art computer vision models relied on hand-

[1]The top-5 error rate is the percentage of images that were not correctly classified into the top 5 most likely classes.

engineered features programmed by human experts, but AlexNet learned all features used to classify the images directly from the data, requiring the humans who programmed it to know essentially nothing about the images themselves. This breakthrough became known as the "AlexNet moment" and marked the beginning of the modern era of machine learning.

This thesis is an attempt to understand the impact of the AlexNet moment on particle physics research in the post-Higgs era. It's worth dwelling very briefly on the fact that both discoveries led to a Nobel Prize in physics, the Higgs for Peter Higgs and François Englert in 2013 and AlexNet for Geoffrey Hinton in 2024. The Nobel Prize for Higgs and Englert was, by my count, the 12th Nobel Prize awarded for the development of the Standard Model[2]. It was the capstone of a very fruitful and now very mature research program. By contrast the Nobel for AlexNet was at least in-part an "instrumentation" award[3], significant for the other discoveries it enables. Particle physics was one of the first scientific disciplines to make use of deep learning. The first papers demonstrating the use of deep learning for event selection [9] and jet tagging [10] were written in 2014, only two years after AlexNet. Since then there has been a remarkable growth in the number of papers applying deep learning to particle physics [11]. This thesis will provide a tour of some of these applications. Concurrently, most other sub-fields of physics have taken interest in applying deep learning tools. Cosmology, astrophysics, gravitational wave science, condensed matter physics, plasma physics, fluid dynamics, quantum information, and optical physics have all seen fruitful applications. There are few areas of active physics research that have not been affected by the AlexNet moment in some sense. While these applications have yet to produce a landmark discovery or new capability like the AlphaFold protein structure prediction model that was awarded the Nobel Prize in chemistry in 2024, the breadth of these applications is a testament to the power of

[2]Notable exclusions from this count are the Nobel Prizes of Feynman, Schwinger, and Tomonaga for the QED theory which is a precursor of the Standard Model, and the Nobel Prizes of Kajita and McDonald for the discovery of neutrino oscillations, which is notably not a part of the Standard Model.

[3]An excellent parallel to AlexNet is Charles Wilson's development of the cloud chamber, which was a tool developed by a meteorologist that became very useful for fundamental physics research and was awarded the Nobel a decade after it was developed.

deep learning for physics research.

In the remainder of the Introduction, I will outline the open problems in fundamental physics[4], and very briefly summarize the last 14 years of progress in machine learning and artificial intelligence. This review serves two purposes. The first is to give context to the very specific research detailed in this thesis. The second is to reflect on the fact that all open problems in particle physics are long-standing, and that the list is essentially unchanged since the 2012 discovery of the Higgs. These problems are extremely subtle and difficult, and many years of effort by very resourceful and intelligent scientists has so far proven insufficient to solve them. By contrast, the field of machine learning and artificial intelligence is completely unrecognizable to itself in 2012. The hope of this thesis, is that some of this dizzying progress can finally break the current impasse facing particle physics.

1.1 Open Problems in Fundamental Physics

The most pressing open problems in fundamental physics can be understood as either the need for a better understanding of Beyond the Standard Model (BSM) physics, or evidence of Beyond ΛCDM cosmology. BSM physics is known to exist through multiple experimental observations that can not be explained by the particle content and forces contained in the Standard Model. However the correct quantum field theory (or other type of theory consistent with the existing Standard Model) that explains these observations is unknown. ΛCDM cosmology accounts for all cosmological observations very well, with the exception of some significant tensions, namely the Hubble tension. My list of the open problems is as follows:

1. **Neutrino oscillations**: Neutrinos are massless under the Standard Model. However many iterations of neutrino experiments have verified that these particles "oscillate" between the three neutrino flavors as they propagate through space. This phenomenon

[4] I generally refer to the fields of particle physics and cosmology as "fundamental physics", given open questions do not necessarily sort neatly into the two sub-fields.

is easily explained if the neutrinos carry a mass that is small enough to remain currently undetected by experiments, but the mechanism that produces this mass is unknown. This fascinating topic is not covered at all in this thesis, but current and next generation neutrino experiments are making active use of deep learning.

2. **Dark matter**: A wide range of astrophysical and cosmological observations, including galaxy rotation curves [12], gravitational lensing, and the large-scale structure of the cosmic microwave background, indicate that roughly 27% of the energy budget of the universe is composed of a non-luminous, non-baryonic form of matter. The Standard Model contains no candidate particle that could account for this "dark matter". Many BSM dark matter candidate particles have been proposed, but none has ever been observed in a particle physics experiment. To the best of our knowledge, dark matter appears to interact with Standard Model particles only through gravity.

3. **Matter–antimatter asymmetry**: The Standard Model predicts that the Big Bang should have produced equal amounts of matter and antimatter, yet the observable universe is composed overwhelmingly of matter. Introduction of this asymmetry requires breaking of the CP discrete symmetry. The known sources of CP violation in the Standard Model—the complex phase in the CKM quark mixing matrix and, to a lesser extent, the strong CP phase discussed below—are far too small to account for the observed asymmetry by many orders of magnitude [13]. A new source of CP violation beyond the Standard Model is therefore required, but remains unknown.

4. **Dark energy and the Hubble tension**: Cosmological observations indicate that the universe is expanding, and that the expansion is speeding up over time. If the universe was composed of only the baryonic matter of the Standard Model and dark matter, this expansion would not be occuring. Therefore there must be an additional type of energy density, termed "dark energy", that is driving this expansion. A good explanation for this dark energy is a non-zero cosmological constant Λ, however this model has signifi-

cant theoretical issues as discussed below. Additionally measurements of the expansion rate of the universe have turned up inconsistencies. The ΛCDM model of cosmology parametrizes the expansion of the universe through the Hubble constant. This constant can be measured through two probes: the cosmic microwave background [14] and local distance-ladder observations [15]. However these two approaches yield values of the Hubble constant H_0 that disagree at the 4–5σ level. This tension could be caused by an unaccounted for systematic effect and is under very active research. However if confirmed this would be a clear sign of beyond ΛCDM cosmology.

5. **Gravity**: The gravitational interactions are not present within the Standard Model. The general theory of relativity provides an extraordinarily accurate classical theory of gravity, but none of the attempts to quantize gravity and incorporate it with the other known forces has received experimental verification. At the energy scales probed by the LHC ($\mathcal{O}$(TeV)) gravitational effects are negligible, so this is not a direct experimental puzzle for collider physics, but our everyday experience of the gravitational force is evidence of BSM physics.

6. **Origin of primordial density fluctuations**: The presence of the Baryon Acoustic Oscillation (BAO) peak in the CMB power spectrum is precisely predicted by ΛCDM cosmology, but requires the presence of primordial density perturbations to seed the oscillations. The mechanism that formed these perturbations in the early universe is not known, though inflation provides a compelling framework which requires the presence of a BSM field. Given there are competing models and experimental evidence is scarce, this is less of a clean example of BSM physics than those listed above, but it is worth mentioning.

In addition to open problems related to experimental observations, there are many more open problems related to the theory of the Standard Model and its possible extensions. To make them more tractable I would group them into three categories. The first, and arguably the most salient at this moment in particle physics, are related to "fine-tunings" of the free parameters of the Standard Model and ΛCDM. Whether these fine-tunings are true problems or only an issue of philosophical priors is often debated, but I find them troubling enough to deserve mention here.

1. **The hierarchy problem**: The Higgs boson mass receives quantum corrections from every particle it couples to. Given it's difficult to imagine a quantum theory of gravity or generally any BSM model that doesn't couple to the Higgs directly or through loops, these quantum corrections should pull the Higgs mass up to the scale of the UV cutoff of the Standard Model. Instead the Higgs has a mass of 125 GeV. This requires either that the quantum corrections are cancelled by some new BSM physics near the electroweak scale, or that the scale of the corrections and the "bare mass" of the Higgs cancel nearly exactly despite their large size. This is the canonical example of a fine-tuning. Supersymmetry, compositeness, and extra dimensions are the most studied solutions to this problem, but none have been confirmed experimentally. The absence of any BSM signal at the LHC has sharpened this "naturalness" tension considerably.

2. **The strong CP problem**: Quantum chromodynamics (QCD) admits a CP-violating term in the Standard Model Lagrangian parameterized by $\bar{\theta}$. Measurements of the neutron electric dipole moment constrain $|\bar{\theta}| \lesssim 10^{-10}$, yet there is no symmetry in the Standard Model that forces this parameter to be small. This is another fine-tuning problem, though less relevant to LHC physics.

3. **The cosmological constant problem**: The observed energy density of the vacuum (dark energy) is roughly 10^{-47} GeV^4, yet naive estimates from quantum field theory predict a value $\sim 10^{120}$ times larger. This is arguably the worst fine-tuning problem in all of physics, and no compelling dynamical mechanism has been found to explain the observed small but non-zero value of the cosmological constant.

The second category consists of problems that are not fine-tunings, at least to the degree of the problems above, but are features of the Standard Model that in some sense "feel odd". Personally I don't discount such feelings as irrelevant as they have produced discoveries in the past[5].

1. **The flavor puzzle**: The Standard Model contains three generations of quarks and leptons that share identical gauge quantum numbers but differ enormously in mass, with Yukawa couplings spanning nearly five orders of magnitude from the electron to the top quark. No symmetry principle explains why there are exactly three generations, why these Yukawa coulings span such large scales, or why the quark mixing angles (CKM matrix) are small while two of the corresponding angles in the lepton sector (PMNS matrix) are large. These hierarchies and mixing patterns are encoded in 13 of the 19 free parameters of the Standard Model. There is no theoretical motivation for their values. Instead they have been determined through experimental measurements, and their somewhat arbitrary structure is an odd feature of the Standard Model.

2. **Accidental symmetries**: Several conservation laws of the Standard Model, baryon number B, and the individual lepton numbers L_e, L_μ, L_τ, are not imposed by the symmetries of the theory. Instead, they are accidents of the Standard Model gauge group, particle content, and renormalizability constraints. The fact that they are present and seem to hold experimentally is a curious feature of the Standard Model.

3. **V-A structure of the weak interaction**: The weak interaction couples exclusively to left-handed fermions, a structure encoded in the $V - A$ (vector minus axial-vector) form of the weak current. Another way of stating this is that the weak interactions are maximally parity violating: the weak force would effect the dynamics of some particles but be completely turned off for their mirror image. While this is experimentally well

[5] Dirac's prediction of the positron through aesthetic considerations about negative solutions to the Dirac equation is perhaps the most relevant one. The GIM mechanism is a particle physics example of how taking a fine-tuning seriously led to new discoveries (the charm quark).

established, there is no theoretical explanation within the Standard Model for why the weak force singles out left-handedness.

The final category is in my opinion the most important one. This is because it is indisputably experimentally and theoretically tracatable at this exact moment in particle physics, and has direct implications for how data from particle physics experiments are processed to hopefully make progress on the many problems quoted above. Furthermore it is the only set of open questions that is directly relevant to the research presented in this thesis. It is problems relating to our ability to *calculate* within the Standard Model.

1. **Calculation of scattering amplitudes**: The Standard Model is predictive because it can be used to calculate scattering amplitudes that are tested every day in particle physics experiments. These amplitudes are almost always calculated at a fixed order in some perturbative expansion, and pushing known amplitudes to ever higher orders is an important open question in particle physics. In particular, improved modeling of Standard Model processes, whether as signal in measurements or background in searches, depends on making progress in this direction.

2. **Non-perturbative QCD**: Quantum chromodynamics is in principle a complete theory of the strong force, but the comparatively large value of the strong coupling constant α_s means perturbative expansions are only valid at high energies. At low energies, lattice QCD provides a systematic non-perturbative approach but is computationally expensive and limited in scope. This leaves us with no first-principles theoretical tools for modeling QCD dynamics at or below the QCD confinement scale of $\Lambda_{QCD} \sim 200$ MeV. Instead we have empirical models whose parameters must be tuned to match to experimental data. Improving these models is an active area of research, for which the cross section measurement presented in Chapter 6 provides valuable experimental data. Apart from this practical problem, the fact that experimental observations like positive mass confined states can't be understood through

perturbation theory is also a deep problem of mathematical physics. Proving that quantum field theories like QCD produce strictly positive mass confined states is one of the Milennium Puzzles of mathematics.

Experimental evidence is needed to guide theoretical progress on both of these problems. It is exactly that evidence that the research in this thesis provides at a uniquely large and flexible scale.

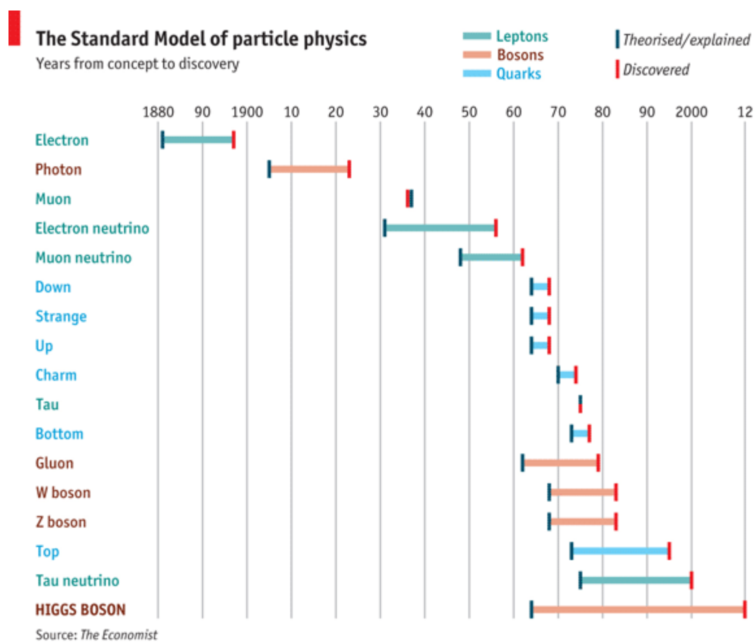

Figure 1.1: A timeline of the theoretical prediction or explanation (green) and experimental discovery (red) of all Standard Model particles. **TODO: cite as fair use**

Finally, I will note that the best way to make progress on the many of these problems is unambiguously to find a new particle (necessarily one not in the Standard Model). Some historical data is relevant here. Figure 1.1 shows a timeline of the theoretical prediction or explanation (green) and experimental discovery (red) of all Standard Model particles. The rate of discoveries between 1965 and 1985, the “golden era” of particle physics, is clearly unique, as is the muon’s experimental discovery before it was conceived of theoretically. However, it is also interesting to examine past droughts. Given the gap between the discovery of the electron and the photon can hardly be considered a time of drought for fundamental physics, the 20 year gap between the discovery of the muon in 1936 and the discovery of the electron neutrino in 1956 strikes me as the longest on record, and will remain so even if the next particle after the Higgs is not discovered for another couple years. Whatever geopolitical forces may have been at play in that time, particle physics research is clearly not always smooth sailing. However the next particle is nearly overdue, and in my opinion prospects for finding it in the next few years are not fantastic. With that sobering fact, I'll contrast a field in which extraordinary progress has happened in the comparative blink of an eye.

1.2 The Deep Learning Revolution

Since the AlexNet moment in 2012, the field of machine learning and artificial intelligence has undergone rapid progress. A brief survey of this progress and how it has entered particle physics research is as follows:

In the years following AlexNet, deeper networks with many millions of parameters provided even better performance on classification tasks [16, 17]. This was an early precursor of the scaling laws that are highly relevant to the current state-of-the-art. The particle physics community moved quickly: the first deep learning jet taggers appeared in 2014 [9, 10], only two years after AlexNet. This period also saw the development of generative neural networks, which produce samples additional samples from the underlying probability distribution of the training data. Generative Adversarial Networks (GANs) [18] and Variational Autoencoders (VAEs) [19] were early generative models that were picked up by the HEP community for generative tasks like fast calorimeter simulation and anomaly detection [20–23]. The most dramatic milestone of this period came in 2016 when AlphaGo [24] defeated Lee Sedol, the world champion Go player. Notably, RL has yet to find widespread application in high-energy physics, but its broader trajectory is relevant to the discussion of agentic AI in the conclusion.

A next landmark was the introduction of the *attention mechanism* and the Transformer architecture in 2017 [25], which demonstrated state-of-the-art performance on machine translation tasks[6]. Attention's remarkable expressivity and the ability to parallelize the required computations on GPU resources meant Transformers replaced recurrent networks as the dominant sequence-processing architecture. The first HEP papers to use attention followed a few years later [26], and attention is now widely used by both ATLAS and CMS for hadronic object identification. Many of the neural networks discussed in this thesis consist of attention backbones.

Following the introduction of attention, the focus in ML/AI research was on scaling and generative modeling. Large language models (LLMs) trained on internet-scale text began to exhibit *emergent* capabilities, which are not present at smaller scales but appear abruptly as model size and training compute increase [27]. The CLIP model [28] demonstrated the ability of neural networks to process "multi-modal" data, for example by developing joint embeddings for text and data simultaneously. During this time the AI + HEP community was broadly focused on generative models. A lot of very rapid progress was driven by applying increasingly elaborate generative model architectures, from GANs to VAEs to normalizing flows [29], to diverse problems like jet generation [30], calorimeter simulation [21], unfolding [31–33], and anomaly detection [34, 35]. Then another landmark occurred between 2021 and 2022 with the introduction of diffusion models [36, 37]. In the AI community, diffusion-based image generators like DALL-E 2, Stable Diffusion, and Midjourney made photorealistic image generation accessible to the general public for the first time. In HEP, diffusion models were quickly picked up and applied to the same set of generative tasks. Diffusion model based unfolding methods are a major topic in this thesis covered in Chapter 5. It is worth pausing to appreciate the distance traveled: neural networks had gone from classifying images to

[6] It was also around this time that Google updated their Google Translate service to use neural machine translation. The development gathered enough media attention that I read about it as a first-year undergraduate. This touched off my interest in machine learning, and my decision to study physics was partially based on the fact that the CMS group was the only research group utilizing deep learning at my undergraduate institution.

generating photorealistic images of arbitrary subjects in the space of 10 years.

Another milestone occurred in late 2023 with the introduction of ChatGPT. The introduction of a reinforcement learning (RL) based post-training [38] to LLMs transformed them from autocomplete engines into conversational assistants capable of writing code, reasoning through multi-step problems, and passing professional licensing examinations such as the bar exam. ChatGPT reached 100 million users in two months, faster than any consumer product in history, and the term "artificial intelligence" entered colloquial use.

In the past two years, frontier LLMs have reached scales of trillions of parameters trained on many terabytes of text data. Simultaneously, advanced RL pipelines have taught language models to use external tools, for example to edit files or search the web. This has produced "agentic" AI systems that are capable of solving increasingly complicated and long-time horizon tasks autonomously. Massive labor market displacement due to these systems is now a real concern, or more relevant to this thesis, some recent physics research papers were written almost entirely by AI systems under supervision from human experts [39, 40]. Regardless of whether this is the future of scientific research, it is evidence of extraordinary progress. I'll return to some opinions on autonomous physics research in the conclusion.

Before moving on, two points about the above deserve some attention. First, although the progress described above is remarkable but it has not been evenly distributed across all of machine learning. Many foundational problems remain as unsolved as they were in 2012. There is still no firm theoretical understanding of why stochastic gradient descent is so effective at training neural networks, and out-of-distribution generalization remains a challenge even on simple tasks. This bears some similarity to how fundamental questions in quantum mechanics were eventually set aside by the physics community because taking the formalism and running with it proved to be much more fruitful. The lesson is that fields move forward in unexpected directions, and that progress sometimes looks like re-focusing the question rather than stubbornly pursuing a solution. Second, the year 2022 strikes me as something of a turning point for the AI + HEP research program. Between 2012 and 2022, the main research directions of the broader AI community produced general-purpose data science tools that the HEP community could adapt relatively directly. Beating benchmarks on HEP tasks essentially required reading the latest NeurIPS proceedings and applying the new methods before other groups did the same. After 2022, the dominant direction of frontier AI research has shifted toward scaling, tool use, and the pursuit of general-purpose intelligence. The question of how pursuit of scaling maps onto fundamental physics is a fascinating one that I'll cover in Chapter 4, but generally speaking this direction does not map as cleanly onto the specific needs of fundamental physics. Without new tools coming out of the frontier AI research community, the AI + HEP community has largely settled on Transformers and diffusion models as the core architectural toolkit. The task is now to use those tools to achieve concrete physics goals.

1.3 Thesis Roadmap

The remainder of this thesis is organized as follows.

Chapter 2 provides the theoretical background in quantum chromodynamics necessary to understand the physics measurement at the heart of this thesis. The treatment is selective. Rather than surveying the Standard Model broadly, it focuses on the aspects of QCD that are directly relevant to jet physics and jet substructure at the LHC. This section will also introduce some of the exotic jet substructure observables measured in Chapter 6.

Chapter 3 introduces the machine learning methods that underpin the research chapters that follow. The goal of this chapter is to bring the reader to a working understanding of the variational latent diffusion models used in the unfolding work of Chapter 5. The chapter is organized as something a progression through the deep learning methods that have been applied within HEP.

Chapter 4 turns to the first of the three original research contributions on top-quark tagging using deep learning methods. It presents the results of an ATLAS study on the training and calibration of top quark taggers with rigorous uncertainty quantification. Given this study was published some time ago the chapter concludes with a summary of recent progress and future directions in jet tagging at the LHC.

Chapter 5 provides the methodological foundation for the cross-section measurement of Chapter 6. It begins with a review of classical unfolding methods, using Iterative Bayesian Unfolding (IBU) as the canonical example. Two AI-based methods are then described in detail: the Variational Latent Diffusion (VLD) model which I developed with collaborators, and Omnifold [41], the method used to perform the cross-section measurement.

Chapter 6 presents the culmination of the thesis: a high and variable dimensional cross section measurement of Z+jets production at the LHC using the Omnifold technique. After describing the methodology, the chapter presents results across four physics use cases that illustrate the unique power of unbinned, high-dimensional unfolding for extracting physics from LHC data.

Chapter 7 concludes the thesis by drawing on overall themes and discussing future roles of machine learning and artificial intelligence in particle physics research.

Chapter 2

Quantum Chromodynamics at the Large Hadron Collider

This chapter provides a very high level overview of quantum chromodynamics (QCD), with a focus on the aspects of the theory relevant to the Large Hadron Collider (LHC). It does not survey the Standard Model or QCD with any completeness. Instead the goal is to motivate two of the observables measured in Chapter 6 from first principles. They will be introduced in Sections 2.2 and 2.3.

QCD is our current best theory of the strong nuclear force. It affects the dynamics of all particles that carry color charge. Another way of putting this is that they transform under a non-trivial representation of the $\mathrm{SU}(3)_c$ sub-group of the Standard Model gauge group:

$$\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \tag{2.1}$$

The particles of the Standard Model are shown in Figure 2.1. The representations of each of the fundamental fields of the Standard Model (grouped as quarks, leptons, gauge bosons, and the Higgs doublet fields) are shown in Table 2.1. Notably all of the fundamental fields

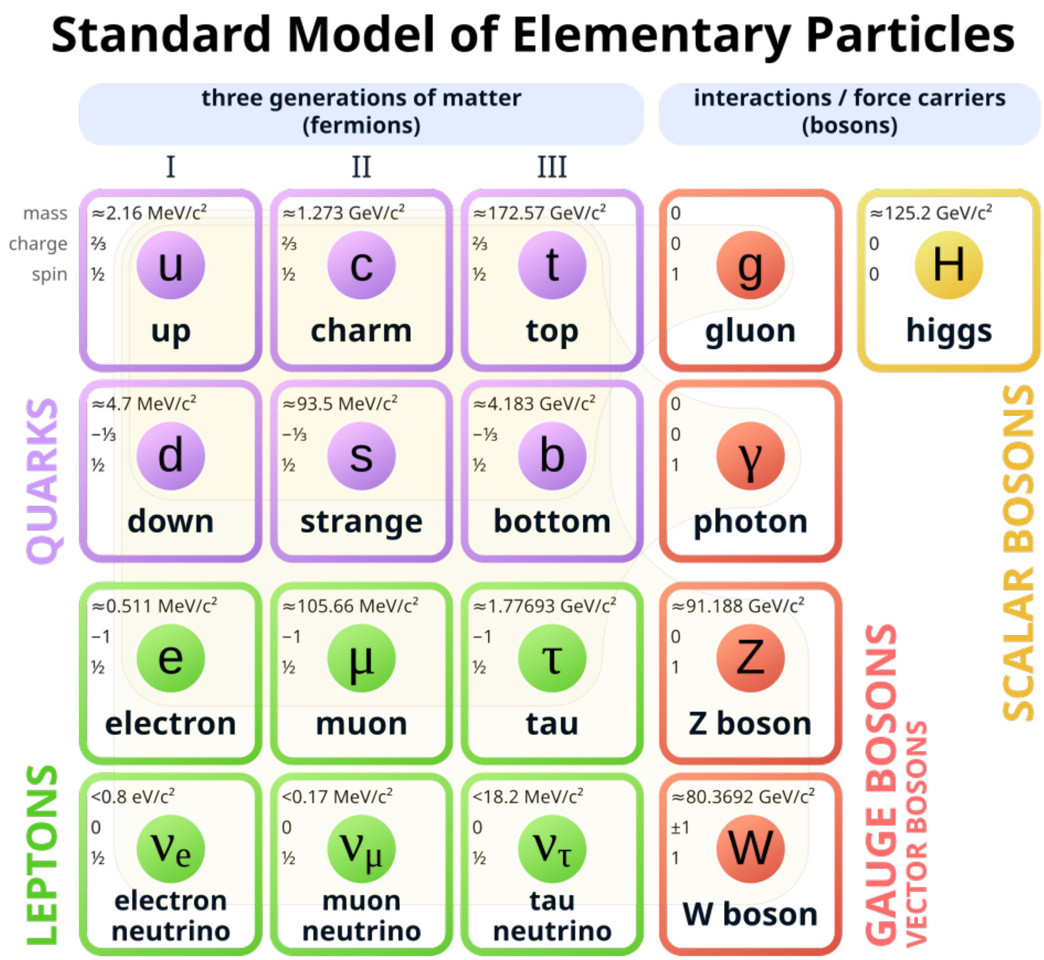

Figure 2.1: The elementary particles of the Standard Model. The dynamics of the 6 quarks and the 8 gluons (all gluon fields are represented by one box in the diagram), described by the theory of quantum chromodynamics, is most relevant to this thesis.

transform in the singlet representation of $\mathrm{SU}(3)_c$ except for the quarks and gluons. These are the particles that carry color charge and participate in strong-force dynamics. The quarks are the massive particles that make up all of the baryonic matter in the universe, and the gluons are the massless gauge bosons that mediate the strong force. These particles are also the input particles to all LHC collisions. The LHC is a hadron collider, meaning it collides color neutral bound states of quarks and gluons. Most notably, the measurement in Chapter 6 is of a cross section measured in proton–proton collisions at $\sqrt{s} = 13$ TeV. Almost all of the LHC collisions recorded by the ATLAS detector contain strong dynamics, making grappling

Table 2.1: Standard Model fields and their representations under the gauge group $SU(3)_c \times SU(2)_L \times U(1)_Y$. Representations are written as $(\mathbf{d_3}, \mathbf{d_2}, Y)$, where d_3 and d_2 are the dimensions of the $SU(3)_c$ and $SU(2)_L$ representations, and Y is the weak hypercharge under the convention $Q = T_3 + Y/2$. Three generations of fermions are implied; all carry identical quantum numbers.

Particle	Symbol	Representation	Spin
Quarks (Left-handed doublets)			
Quark doublet	$Q_L = \binom{u}{d}_L$	$(\mathbf{3}, \mathbf{2}, +\frac{1}{3})$	1/2
Quarks (Right-handed singlets)			
Up-type quark	u_R	$(\mathbf{3}, \mathbf{1}, +\frac{4}{3})$	1/2
Down-type quark	d_R	$(\mathbf{3}, \mathbf{1}, -\frac{2}{3})$	1/2
Leptons (Left-handed doublets)			
Lepton doublet	$L_L = \binom{\nu_e}{e}_L$	$(\mathbf{1}, \mathbf{2}, -1)$	1/2
Leptons (Right-handed singlets)			
Charged lepton	e_R	$(\mathbf{1}, \mathbf{1}, -2)$	1/2
Gauge Bosons			
Gluons	g^a_μ	$(\mathbf{8}, \mathbf{1}, 0)$	1
W bosons	W^i_μ	$(\mathbf{1}, \mathbf{3}, 0)$	1
B boson	B_μ	$(\mathbf{1}, \mathbf{1}, 0)$	1
Higgs Sector			
Higgs doublet	H	$(\mathbf{1}, \mathbf{2}, +1)$	0

with the complexities of this force a central challenge of LHC physics. In particular, the challenge is handling *jets*, which are the experimental signature of the production of high transverse momentum (p_T) color-charged particles. Jets will be introduced in Section 2.1, and two methods for studying them will be introduced in Sections 2.2 and 2.3.

2.1 Jets and Jet Substructure at the LHC

2.1.1 The QCD Lagrangian and Asymptotic Freedom

The dynamics of quarks and gluons are governed by the QCD Lagrangian,

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4} G^a_{\mu\nu} G_a^{\mu\nu} + \sum_f \bar{q}_f \left(i\gamma^\mu D_\mu - m_f \right) q_f, \tag{2.2}$$

where the sum runs over the six quark flavors f, $D_\mu = \partial_\mu - i g_s T^a A^a_\mu$ is the covariant derivative, g_s is the strong coupling constant, and T^a are the generators of SU(3) in the fundamental representation. The gluon field strength tensor is

$$G^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g_s f^{abc} A^b_\mu A^c_\nu, \tag{2.3}$$

where f^{abc} are the SU(3) structure constants and A^a_μ are the gluon fields.

A critical distinction between QCD and QED is that the structure constants f^{abc} in Equation 2.3 are non-zero for the SU(3) gauge group of QCD. The presence of non-zero structure constants introduces generates cubic and quartic gluon self-interaction vertices in the Feynman rules, allowing gluons to couple directly to each other, as shown in Figure 2.2. Another way of putting this is that the gluon field itself carries color charge. In QED, the photon carries no electric charge and no analogous photon self-coupling exists.

(a) Three-gluon vertex ($\sim g_s f^{abc}$)

(b) Four-gluon vertex ($\sim g_s^2 f^{abe} f^{cde}$)

Figure 2.2: The gluon self-interaction vertices that arise from the non-Abelian structure of QCD.

These self-coupling diagrams have important consequences for the running of the strong coupling constant g_s. For the electromagnetic interactions, the coefficient of the one-loop beta function is positive, meaning the coupling increases with the energy scale. For the strong interactions, the gluon self-interaction contributes a large negative term to the one-loop beta function,

$$\mu \frac{dg_s}{d\mu} = -\frac{g_s^3}{16\pi^2}\left(\frac{11}{3}C_A - \frac{4}{3}T_F n_f\right) + \mathcal{O}(g_s^4), \tag{2.4}$$

where $C_A = 3$ is the adjoint Casimir, $T_F = 1/2$, and n_f is the number of active quark flavors. The first term in the parentheses is the important one that arises from the gluon-gluon self interaction. For $n_f \leq 16$ the quantity in the parentheses is positive, making the overall sign negative, so g_s decreases as the energy scale μ increases. This is in contrast to the electromagnetic interactions whose strength *increases* with energy scale. The running of the QCD coupling constant, verified by many experimental measurements, is illustrated in Figure 2.3. At high energy scales Q, the strong coupling (here plotting α_s rather than g_s)

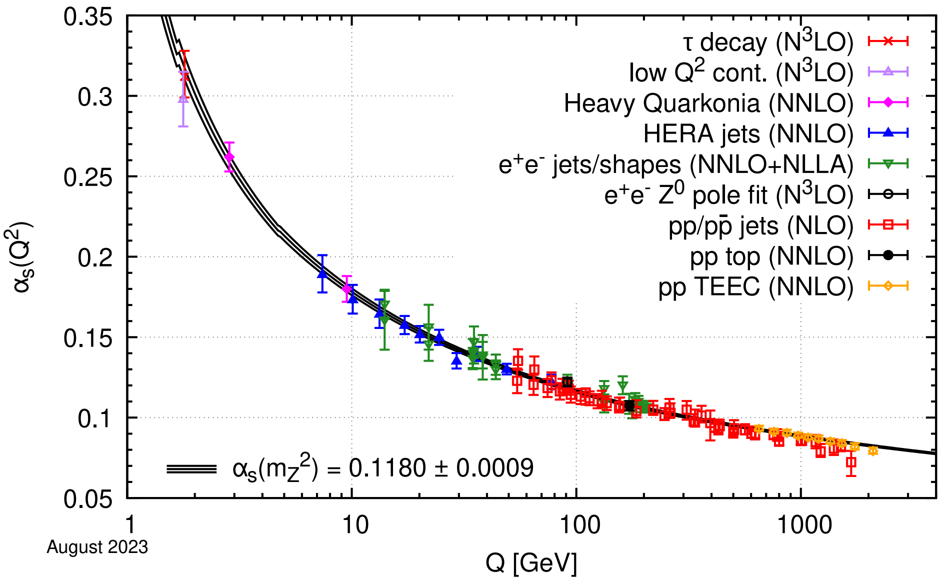

Figure 2.3: A compilation of experimental measurements of the strong coupling α_s as a function of the energy scale Q, illustrating asymptotic freedom through the decrease of the coupling at high energies. **TODO: cite PDG**

is less than 0.1. This is the perturbative QCD regime in which a perturbative expansion in powers of the strong coupling is well defined. Here the strong force dynamics are similar to the EM dynamics and color charged particles can freely propagate. At low energy scales the coupling begins to diverge. Below the QCD hadronization scale $\Lambda_{\mathrm{QCD}} \sim 200$ MeV, the strong force is so powerful that it becomes energetically favorable for color-charged particles to pull opposite color-charged particles out of the vacuum such that the overall color charge is neutralized. This is known as *hadronization*, since color charged particles like quarks and gluons become bound up in composite particles known as hadrons. The protons and neutrons inside atomic nuclei are the most important hadrons, as they make up the vast majority of matter in the universe. Note that this hadronization is an inherently non-perturbative process since the strong coupling constant is large by assumption. The decreasing strength of the strong force with energy scale is known as asymptotic freedom[42, 43]. It offers a compelling example of how an energy scale (namely Λ_{QCD}) can be generated by the structure of a theory rather than being put in by free parameters as is the case for many of the theoretical issues described in Chapter 1.

2.1.2 Parton Showers and Infrared and Collinear Divergences

At high energies, quarks and gluons can propagate freely but this does not mean that the propagation is simple. The QCD splitting functions, which describe the probability of a parton[1] to emit a gluon, have soft and collinear divergences. The leading-order splitting functions given schematically by

$$dP = \frac{\alpha_s}{\pi} C_f \frac{d\epsilon}{\epsilon} \frac{d\theta^2}{\theta^2} \tag{2.5}$$

where $\epsilon = 1 - z$ is one minus the momentum fraction carried by one of the daughter partons z, $C_F = 4/3$ is the fundamental Casimir of SU(3), and θ is the emission angle between the daughter partons. This splitting function diverges as either ϵ or θ become small, correspond-

[1]Quarks and gluons are often referred to as partons, since they are the constituent parts of the hadrons we observe at low energies. At high energies, the term is still used but is less descriptive since quarks and gluons can propagate freely.

ing to the emission of a soft or collinear gluon by the parent parton[2]. These divergences exist whether the parent parton is a quark or a gluon, so gluons emitted by an initiating parton can in-turn emit more gluons. This cascade of largely soft and collinear gluons is known as a *parton shower*, which continues until all of the emitted partons reach energies near Λ_{QCD}.

The practical consequence is that any quark or gluon produced in an LHC collision immediately undergoes a parton shower before it can interact with a detector. The collimation of this shower (all the resulting quarks and gluons travel in roughly the same direction) follows directly from the collinear divergence of the splitting functions. Emissions at large angles are heavily suppressed.

2.1.3 Jets

These two sequential processes, parton showering at high energies and hadronization at low energies, imply that the result of the production of a high $p_T$quark or gluon in an LHC collision is a collimated spray of hadrons that can interact with a detector. This signature is known as a *jet*. Measurement of a jet requires that the four-vector of each hadron within the jet, typically called *jet constituents*, be measured by the detector. The four-vector of the jet itself can then be calculated by summing the four-vectors of the jet constituents, and then the four-vector of the jet can be used as a stand-in for the four-vector of the initiating parton when analyzing LHC collisions. This simple prescription is made much more complicated by a few experimental realities:

1. Choosing which measured hadrons to include in a jet is non-trivial. In proton-proton collisions, the presence of multiple jets, multiple parton interactions, and initial and final state radiation significantly complicate this question. Studying jets typically requires specifying a *jet clustering algorithm*, which leads to many experimental and

[2] The splitting $g \to qq$ does not carry divergences, meaning that this "flavor excitation" is a small effect in the shower. However its effect does become important when α_s becomes large at small energy scales. In particular it plays a roll in hadronization.

theoretical subtleties.

2. The general purpose detectors on the LHC, ATLAS and CMS, are not sensitive to the entire four-vector for an arbitrary hadron. The procedure is covered in some detail in Chapter 6, but for now suffice to say that some details about the underlying hadrons are lost. Significant approximations are made to put these details back in so that the four-vector of the jet can be calculated.

3. Even the parts of the four-vector that are measured are not measured with perfect resolution. Some *smearing* of the true kinematic quantities is inevitable, and can cause significant distortions in the four-vector of the jet.

4. The LHC operates with many proton-proton collisions per bunch crossing. The presence of these *pileup collisions* superimposed on top of the collision of interest can contaminate the jet, essentially adding hadrons to it that were not produced by the shower initiated by the parton.

Each of these points adds significant complications to the experimental study of jets, even if the only goal is to infer the four-vector of the initiating parton. However there is a huge amount of interesting and unknown physics contained in the parton shower and hadronization. This physics can be probed through examining the underlying measurements of the jet constituents. Studying this *jet substructure* requires processing the full, high-dimensional data type that describes all hadrons within a jet. The complexity and high dimensionality of these data types makes them natural applications for deep learning methods. This is why the jet substructure community saw the earliest and most enthusiastic adoption of deep learning in LHC physics [44, 45].

The study of jet substructure is organized around substructure observables. These are quantities that can be calculated using the four-vectors of the jet constituents that capture certain features of a jet. Some example observables are discussed in Chapter 4. As is the case for all physics, the study of jet substructure moves forward through an interplay between theorists and experimentalists. Theorists make predictions of jet substructure observables within the framework of QCD, and experimentalists make measurements of these observables that can be compared to and used to refine the theory. As Chapter 6 is a deep dive into the experimental side of this program, some brief description of the theory side is warranted here.

There are two approaches to calculating jet substructure observables within QCD. The first approach is to directly calculate a given observable within perturbative QCD. A complete calculation involves three ingredients [44]: (1) a fixed-order calculation in α_s capturing hard emissions, (2) resummation of the large logarithms arising from soft and collinear emissions, and (3) modeling of non-perturbative effects, typically through a shape function that accounts for hadronization corrections. The result is a systematically improvable prediction for the observable's cross section [46].

The second approach is to recursively apply the splitting function in Equation 2.5. Given the probability of a splitting at a given scale does not depend on splittings at any previous scales, the chain of splittings can be described as a Markov chain, and samples from the probability distribution over the final configuration of the parton shower can be obtained by Markov Chain Monte Carlo (MCMC) methods. Samples provided by the MCMC are a fully exclusive partonic final state, essentially a list of partons at or above the QCD hadronization scale Λ_{QCD}. In order to compare to experimental measurements, phenomenological models of hadronization can be applied [47, 48]. Unlike the parton shower sampling, these models are not based on first principles QCD but are models with free parameters that must be *tuned* to match experimental data. The combination of sampling parton shower configurations through MCMC followed by application of a hadronization model gives rise to "Monte Carlo" event generators which are capable of simulation proton-proton collisions with high accuracy [49–51]. This approach is fully general-purpose: it can provide predictions for any observable, including those that are too complicated for analytic resummation. The limitation is that the accuracy of parton shower predictions is harder to characterize systematically, and assigned theory uncertainties are less rigorous than those from resummed calculations.

An important practical distinction between these two approaches is the form of their output. Resummed calculations provide smooth analytic predictions for inclusive cross sections, while event generators produce discrete lists of outgoing hadron four-momenta. Only the latter can be passed through a detector simulation. Given highly accurate simulations of a particle's interaction with matter exist, the detector can be simulated with great accuracy [52, 53]. Chaining together a fixed order calculation (either in QCD or the SM more generally), parton shower, hadronization model, and detector simulation defines a *simulation chain* capable of producing simulated collision events that resemble the data collected by the experiments at the LHC. This ability explains why event generators are used copiously in essentially every analysis of LHC data, while resummed calculations are only relevant within the sub-field of jet substructure and QCD. This difference between the two approaches is also a subtle but important motivation for the unfolding methods discussed in Chapter 5.

The remainder of this chapter discusses two jet substructure observables which are measured and compared to theoretical predictions in Chapter 6: energy-energy correlators and the intrinsic dimensionality of jets. The reason for exploring these observables in particular is that they are very difficult to unfold with conventional binned methods. Detailed discussion of this will be delayed until Chapter 5, but some introduction to the observables from a theoretical standpoint is due first.

2.2 Energy correlator observables

2.2.1 Definition and History

Energy correlators are a class of collider observables with a long history. They were first introduced as event-shape observables in the context of electron-positron collisions [54–57], where they were used to study the radiation pattern of hadronic final states. One important application is the extraction of the strong coupling constant α_s. Several of the measurements shown in green in Figure 2.3 are derived from energy correlator measurements in e^+e^- collisions.

The n-point energy correlator (EnC) is defined as the energy-weighted n-point correlation function of the radiation pattern produced by a particle collision. This mathematical object is also used in cosmology to study the CMB and in condensed matter physics to study phase transitions and criticality[3]. The simplest energy correlator is the two-point energy correlator (E2C or EEC). Measuring this quantity involves building a histogram of the opening angle between objects produced by a collision, with the histogram entries weighted by the product of the energies of the two objects. The energy weighting ensures that the observables are infrared and collinear (IRC) safe. Either individual particles or clustered jets can serve as inputs [60–63]. Most studies have focused on the properties of the EEC, but high order correlators are also of some theoretical interest. For high order correlators the number of pairs (or tuples) of objects grows combinatorially with the particle multiplicity, which makes the measurement substantially more computationally demanding [64].

2.2.2 Energy Correlators as Jet Substructure Observables

The use of energy correlators specifically as jet substructure observables is a more recent development. See Ref. [66] for a recent review. Interest has been driven by a few properties

[3] The connection to the operator product expansion and light-ray operators in quantum field theory makes this analogy more than superficial; see [58, 59] for a formal treatment.

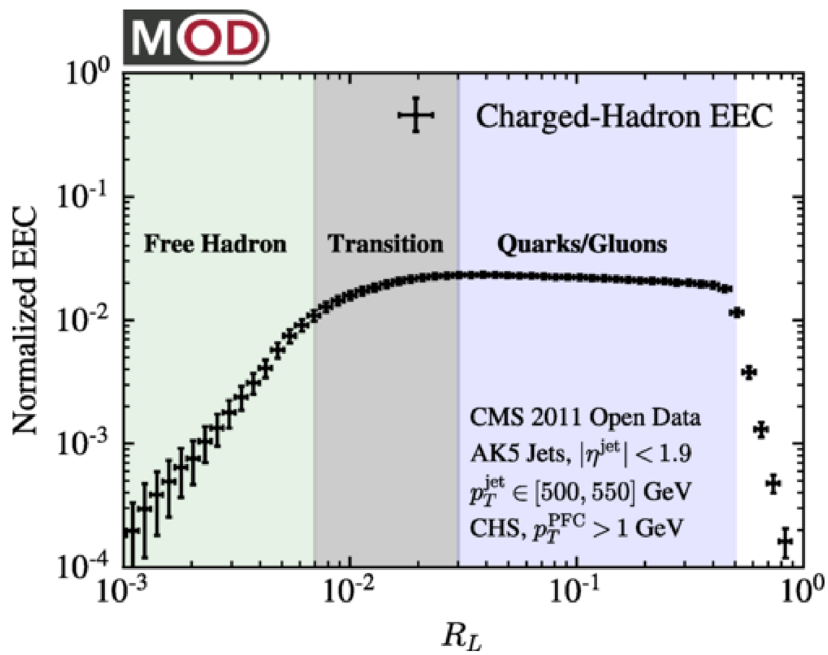

Figure 2.4: The two-point energy correlator measured in CMS 2011 open data for high $p_T$jets, showing distinct free-hadron, transition, and quark/gluon regimes as a function of the angular scale R_L (called z in the text). Figure adapted from Ref. [65].

of the two-point correlator in the collinear limit that can be probed using the particles within jets. The two-point correlator is given as

$$EEC(z) = \frac{1}{\sigma} \sum_{i,j} \int d\sigma \frac{E_i E_j}{Q^2} \delta(z - \frac{1 - \cos\theta_{ij}}{2}) \tag{2.6}$$

where σ is the total cross section, E_i is the transverse energy of the i-th particle, Q is the sum of the transverse energies of all particles ($\sum_i E_i$), θ_{ij} is the angle between the three-momentum vectors of the i-th and j-th particles, and $z = \frac{1-\cos\theta_{ij}}{2}$. The first property is that in perturbative QCD at high energies, the scaling behavior of the EEC as a function of the opening angle z between freely propagating partons is predicted to follow a power law. The exponent of this power law is related to a formal field theoretical quantity known as the cusp anomalous dimension. This provides a direct connection between a measurable quantity and a fundamental property of QCD. In general, the larger the angular separation between particles z, the earlier in the jet formation process (or the higher the energy) is being probed. This means that are moderate values of z, where particles are contained within a jet but last interacted with each other early in the jet formation process, a power law scaling in the EEC should be observed.

Second, deviations from this perturbative power-law scaling encode non-perturbative effects like hadronization. As z grows smaller, the EEC quantifies correlations between increasingly collinear particles, meaning they were emitted later in the jet formation process. Therefore the very small z regime should be sensitive to dynamics below the scale Λ_{QCD} where hadronization occurs. Hadrons are color neutral and do not interact by the strong force, and so the correlation should reflect non-interacting particles. Namely it should have a linear dependance on the angular separation.

The combination of these two properties mean that two distinct scaling regimes should be visible in the EEC: a power law corresponding to freely propagating partons at high energies at high z, and a linear scaling corresponding to non-interacting hadrons at low z. Ref. [65] first illustrated these two distinct scaling regimes using CMS open data, as shown in Figure 2.4. The two scaling regimes are shaded in blue and green respectively, with a transition regime corresponding to the hadronization process linking them. The scaling regimes are very well understood from a theoretical standpoint, but the transition between them is intractable given our current theoretical tools, motivating it as a useful object to measure experimentally. Observing this transition cleanly requires jets with high transverse momentum and large particle multiplicity, so that the collinear regime is well-separated from the hadronization scale. The LHC produces such jets copiously. This was not the case at LEP, where the available center-of-mass energy limited the jet p_T, which is why the energy correlators were relatively unexplored as jet substructure tools in the e^+e^- era.

The final property is that the energy correlators are among the most theoretically tractable jet substructure observables from the perspective of higher-order perturbative calculations. The EEC in e^+e^- collisions has been computed to next-to-next-to-next-to-leading order

(N^3LO), which is a very high level of accuracy for a jet substructure quantity [67]. Recent work has extended high-order calculations to the EEC in proton-proton collisions as well [68].

2.2.3 Experimental Status

The theoretical properties described above have made energy correlators a rapidly developing experimental topic. Several measurements have been made in recent years, though significant gaps remain. I only summarize here energy correlator measurements using particles, rather than jets, since this is needed to access the collinear regime.

The most directly relevant existing measurement is the E2C and E3C measurement by the CMS collaboration in proton-proton collisions at the LHC, which also extracted a value of α_s [63]. This is the only published measurement using high $p_T$jets from pp collisions at the LHC and serves as a primary point of comparison for EEC application of the measurement presented in this thesis. The ALICE collaboration additionally measured the EEC in charm-tagged jets in proton-proton collisions, but at a lower center-of-mass energy and with less luminosity [69]. Energy correlator measurements have also been carried out in heavy-ion [70, 71] and proton-ion [72] collisions at the LHC. These measurements feature lower $p_T$jets, where the observed EEC distributions are modified due to interaction with the quark-gluon plasma. The STAR collaboration has measured the EEC at the lower center-of-mass energies available at RHIC [73]. Legacy analyses have reprocessed archived data from the ALEPH experiment at LEP [74], where the EEC is measured with all tracks given the substantially lower particle multiplicity produced by electron-positron collisions. Finally, the H1 collaboration has presented preliminary results measuring the EEC distribution in electron-proton deep inelastic scattering events using all tracks [75]. This result is particularly relevant because it uses the same approach to unfolding as the measurement in this thesis. It will be discussed in detail in Chapter 5. Table 2.2 summarizes the experimental landscape.

Table 2.2: Summary of published experimental measurements of energy-energy correlator observables using particles. Jet transverse-momentum ranges are approximate and refer to the leading-jet or inclusive-jet selection; a dash (–) indicates measurements performed on all final-state particles without jet clustering. The asterisk denotes measurements made with charged-particle jets rather than jets that include the neutral component. The dagger denotes preliminary results.

Experiment	System	$\sqrt{s}$	Observable	Jet p_T range
ALEPH [74]	e^+e^-	91.2 GeV	E2C all tracks	–
CMS [63]	pp	13 TeV	E2C, E3C inside jets	500–2500 GeV
ALICE [69]	pp	13 TeV	E2C inside jets	10–30 GeV*
ALICE [70]	pp	5.02 TeV	E2C inside jets	20–80 GeV*
STAR [73]	pp	200 GeV	E2C inside jets	25–55 GeV
ALICE† [72]	pPb and pp	5.02 TeV	E2C, E3C inside jets	20–80 GeV*
CMS [71]	PbPb	5.02 TeV	E2C inside jets	120–200 GeV
H1† [75]	ep	320 GeV	E2C all tracks	–

Despite this progress, the existing suite of measurements has notable gaps. First, there is only one measurement from CMS which uses the high $p_T$jets provided by the LHC to measure the EEC well above the confinement scale [63]. A similar measurement performed by ATLAS would be a valuable contribution. Second, all current LHC measurements cluster particles into jets before performing the correlator measurement on the jet substructure. This procedure is experimentally convenient, particularly for the unfolding step discussed in Chapter 5, but it introduces additional theoretical complications associated with the jet clustering. A measurement performed on all particles in the event, without prior jet clustering, would allow measurement of the entire perturbative scaling regime without the distribution being cut-off due to the jet radius. Third, all existing measurements use only charged particles, since neutral particles such as photons and neutral pions are measured with substantially worse angular resolution. Theoretical predictions for charged-particle-only measurements require the track function formalism [76, 77], which introduces an additional layer of non-perturbative modeling. Discussion of this is beyond the scope here, but work on this formalism and associated measurements is on-going [78]. One of the primary physics contributions of the measurement presented in this thesis is to address the first two gaps.

2.3 Intrinsic Dimensionality of Jets

The energy-energy correlators discussed in Section 2.2 are jet substructure observables with a direct connection to the formal structure of QCD. In contrast, the intrinsic dimensionality of jets is an observable motivated by a basic question about the structure of the data themselves rather than ties to the underlying theory. This is a very new observable that has only been treated, to my knowledge, in a few research papers [79, 80]. There are likely interesting connections to the underlying theory. For example the scale dependence of the intrinsic dimension (see below) could in principle encode information about the parton shower branching structure, but this remains an essentially unexplored topic. Part of the motivation for measuring this observable is to demonstrate that measurements of it are possible, hopefully motivating further theoretical work. The rest of the motivation is to demonstrate that deep learning methods allow us to ask questions about the data in its native, high dimensionality.

2.3.1 The Manifold Hypothesis and Intrinsic Dimension

The manifold hypothesis states that high-dimensional data generated by natural processes tend to cluster on lower-dimensional manifolds embedded in the high-dimensional space. Put another way, the naive high dimensional representation we use to describe the data is redundant. Fewer coordinates are needed to describe the data than the naive dimensionality would suggest, demonstrating that there is structure in the high dimensional space that is invisible to humans examining projections (e.g. histograms or scatter plots) but easily identifiable for deep learning models. This "hypothesis" is not really a hypothesis but an empirical observation about high dimensional data across many domains. This observation is implicitly known in particle physics, given our data is known to be generated by relatively simple processes and respect certain symmetries, but it has not been explicitly quantified.

The intrinsic dimension (ID) of a dataset is the dimension of the underlying manifold. Put a third way, the manifold hypothesis states that the intrinsic dimension of a dataset is typically

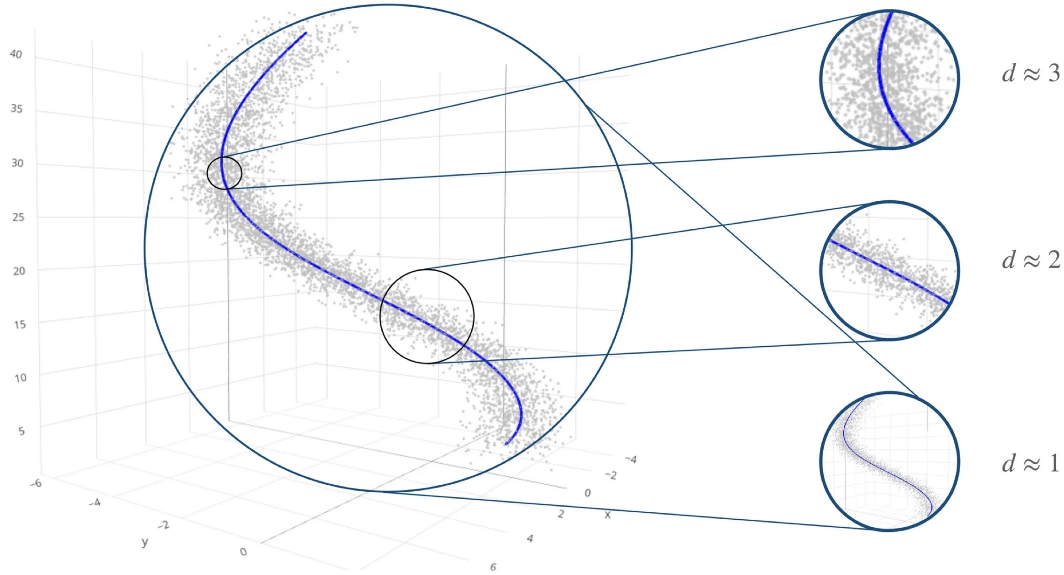

Figure 2.5: Illustration of the scale dependence of the intrinsic dimension. At large length scales the data appear as a single point (zero dimensions); at smaller scales the geometry of the underlying data manifold becomes visible and the estimated dimension increases. **TODO: cite!**

much less than the number of naive coordinates we use to describe the data. An important feature of the ID is that it is *scale-dependent*. Figure 2.5 provides a visual illustration of this scale dependence. The data cluster on a manifold embedded in the naive three-dimensional coordinates used to describe the data, but the apparent dimension of the manifold depends on the length scale. At large distances the manifold looks like a curve that can be parametrized with one free parameter, so the dimension is one. At smaller distances, the noise in the data gets resolved and the manifold appears first two- and then three-dimensional. Clearly the intrinsic dimension also does not need to be an integer. Fractal dimensions are expected as interpolations betwen the dimensions illustrated in the Figure.

A variety of estimators have been developed to compute the intrinsic dimension of a dataset [81–85]. A quick summary is as follows:

1. Principal component analysis (PCA) methods count the number of statistically significant eigenvalues of the data covariance matrix. They assume the data lie on a linear

subspace, which limits their applicability to curved manifolds [81].

2. Fractal and correlation dimension methods exploit the scaling of the number of data points contained within a ball of radius r as r varies. The correlation dimension [82] is the most widely used estimator in this family. It was the first estimator to be utilized as a jet substructure observable[79].

3. Nearest-neighbor maximum likelihood methods apply a maximum likelihood estimator to the distribution of distances between neighboring data points [83]. This is currently the most widely used class of intrinsic dimension estimator in machine learning, and it is the approach used in this thesis. It was recently proposed as a test statistic for searches for physics beyond the Standard Model at the LHC [80].

4. A fourth family, based on concentration-of-measure phenomena in high-dimensional spaces, has been developed but not yet applied to particle physics data [84, 85].

2.3.2 The Energy Mover's Distance

Most intrinsic dimension estimators require that the vector space used to describe the data be equipped with a metric so that pairwise distances between points can be calculated. For jet data, the natural choice is the Energy Mover's Distance (EMD), introduced in [79]. The EMD between two jets $\mathcal{E}$ and $\mathcal{E}'$ is defined as the minimum amount of "work" required to rearrange the energy flow of one jet into the other,

$$EMD(\mathcal{E}_k, \mathcal{E}_l) = \min_{f_{ij}} \sum_{i,j} f_{ij} \frac{\theta_{ij}}{R} + |\sum_i E_i - \sum_j E_j|, \tag{2.7}$$

where f_{ij} is the optimal transport plan moving energy from particle i in one jet to particle j in the other, θ_{ij} is the angle between the three-momentum vectors of the i-th and j-th particles, and R is the radius of the jet. The second term accounts for differences in the overall energy normalization between the jets. The f_{ij} satisfy the following constraints:

$$f_{ij} \geq 0, \tag{2.8}$$

$$\sum_j f_{ij} \leq E_i, \tag{2.9}$$

$$\sum_i f_{ij} \leq E_j, \tag{2.10}$$

$$\sum_{i,j} f_{ij} = E_{min}, \tag{2.11}$$

Given the cost function for the EMD is weighted by the particle energies, the EMD is infrared and collinear safe. This makes it a physically well-motivated metric for jet data. To use the EMD as a substructure observable, it is useful to preprocess the jets to remove any irrelevant features, such as the location of the jet in the pseudorapidity-azimuth (η-ϕ) plane or its rotation. The standard approach, used throughout, is to apply a lorentz boost and rotation to the jets such that they are centered in the η-ϕ plane, and then rotate the jets about their axis such that the first principle component of the jet radiation pattern points toward the positive ϕ axis. See Chapter 6 for more details.

2.3.3 The Nearest-Neighbor Intrinsic Dimension Estimator

Given a set of N jets with EMDs calculated between all pairs (the number of EMDs to compute is N^2), the nearest-neighbor intrinsic dimension (NNID) estimator [80, 83] constructs a likelihood from the ratios of distances to successive nearest neighbors. For a jet k, define the distance ratio

$$\mu_{k,ij} = \frac{\mathrm{EMD}(k, \mathrm{NN}_j(k))}{\mathrm{EMD}(k, \mathrm{NN}_i(k))}, \tag{2.12}$$

where $\mathrm{NN}_i(k)$ denotes the ith nearest neighbor of jet k in the EMD metric. These distance ratios depend on the values of the unphysical indices i and j. In what follows, I follow

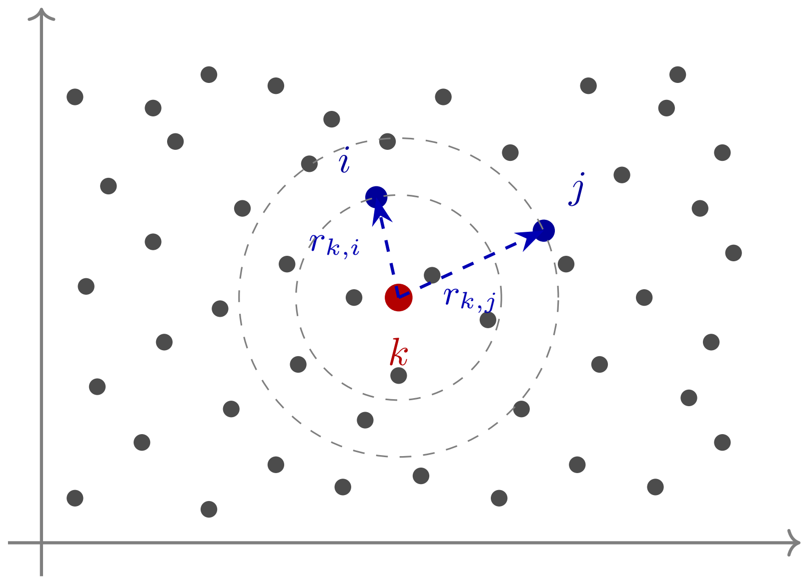

Figure 2.6: Illustration of the NNID estimator in a simple two-dimensional space. For a reference point (center), the distances $r_{k,i}$ to the ith nearest neighbor and $r_{k,j}$ to the jth nearest neighbor define the ratio $\mu_{k,ij}$. In this particular case $i = 5$ and $j = 10$. The distribution of these ratios across many reference points encodes the intrinsic dimension of the data manifold.

Ref. [80] and set $j = 2i$ throughout, reducing the number of free parameters to one. An illustration of these ratios is provided for two dimensional data in Figure 2.6.

In a d-dimensional space, the ratio $\mu_{k,ij}$ follows a Pareto distribution whose shape parameter encodes d. The intrinsic dimension for a given choice of i and $j = 2i$ can then be estimated by minimizing the negative log-likelihood,

$$\begin{aligned} L &= -\sum_{k=1}^{N} \log f(\mu_{k,ij}; d) \\ &= -N \log d + (1 + id) \sum_{k=1}^{N} \log \mu_{k,ij} - (j - i - 1) \sum_{k=1}^{N} \log \left(1 - \mu_{k,ij}^{d}\right) . \end{aligned} \tag{2.13}$$

$f(\mu_{k,ij}; d)$ is the Pareto probability density evaluated at the ratio for jet k, which has an analytic form written in the second line. Numerically minimizing Equation 2.13 over d with all jets k included in the sum yields the estimated intrinsic dimension.

To attach a physical meaning to the index i, one can compute the median EMD between a jet and its ith nearest neighbor across the dataset. Plotting the NNID estimate against this median EMD produces a curve showing how the apparent dimensionality of the jet phase space varies with the typical EMD between the jet-pairs in the denominator of Equation 2.12. In practice, the calculation must also accommodate per-event weights, which arise from the event generator used to produce samples of simulated events, and the unfolding procedure discussed in Chapter 5. To do this each jet's contribution to the likelihood is scaled by its weight, and the nearest-neighbor counting is replaced by its weighted analogue.

2.3.4 Generator-Level Results

Given the novelty of this observable in particle physics, it is worth presenting some generator level results before moving on. Jets clustered from particle-level charged particles are sampled from the MadGraph and Sherpa samples described in Chapter 6. See (**TODO: make foward reference better**) for more details. Each point in the curves corresponds to a given choice of the unphysical index i, which produces a NNID and median EMD estimate. The uncertainties on the theoretical predictions are shown as colored boxes. The results are shown in three panels covering different regions of the NNID vs median EMD curve in order to resolve the fine structure.

The two generators generally agree but show visible differences at very low EMD values. The NNID at the lowest scales probes fine differences in jet substructure between two jets that globally look very similar. It is not surprising that these subtle differences are sensitive to imperfections in the event generators.

The first experimental measurement of the intrinsic dimensionality of jets is presented in this thesis. It represents a somewhat different approach to jet physics, in which the focus is on asking basic questions about the experimental data "as data" and then evaluating the ability of QCD and the Standard Model to reproduce the answers. This experimentally driven approach to measurements complements the typical approach of targeting observables due to their connection to the underlying theory.

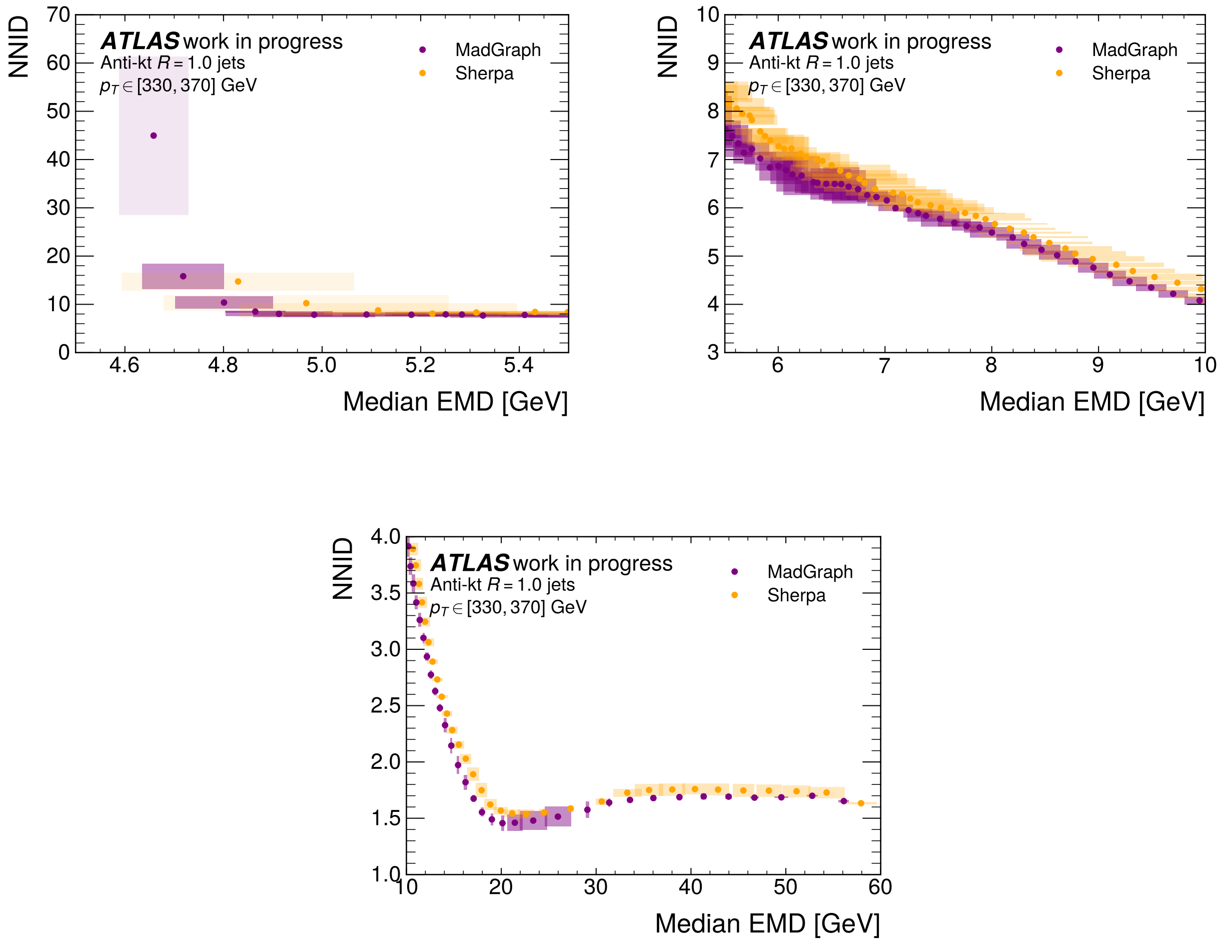

Figure 2.7: Nearest-neighbor intrinsic dimension (NNID) as a function of the median EMD scale, computed at truth level for the MadGraph (purple) and Sherpa (gold) event generators used in Chapter 6. The three panels show the same curve over different ranges of the EMD axis to resolve fine structure. Differences between the generators are most pronounced at very low EMD scales.

Chapter 3

Machine Learning and Artificial Intelligence

This Chapter serves as a general introduction to the deep learning methods used throughout the remainder of the thesis[1]. Like Chapter 2, the goal is not to be thorough. Instead only the information needed to understand the particular deep learning methods applied to particle physics data in this thesis will be covered.

3.1 Overview: ML as a Paradigm Shift

Machine learning is the idea that complex rules or behaviors can be learned directly from data by optimization algorithms, rather than programmed explicitly by human experts. In classical data analysis, a domain expert encodes their knowledge as a fixed set of rules that are then applied to data to extract a quantitative summary of the information. It is worth emphasizing that this approach often works very well. Particle physicists have become very adept at choosing relevant features of the data and learning from them. However this

[1] Note that this can be thought of as the "instrumentation" chapter of this thesis, which might otherwise be dedicated to information on the ATLAS detector.

approach is rarely optimal, especially in cases where the optimal solution exists but is too complex to specify by hand. To draw an example from Chapter 4, no expert-designed feature set captures the correct decision-surface for classifying jets initiated by boosted top quarks from the light quark and gluon jet background. This is due to a lack of knowledge about certain features of the jet formation process (e.g. hadronization), but also just due to complexity. Specifying the optimal feature by hand is probably not possible, but machine learning methods allow this complexity to be learned from the data.

Machine learning methods organize naturally into several paradigms based on the structure of the available training information:

- *Supervised learning* trains a model using labeled examples, where each input is paired with a target output. Classification and regression are the canonical supervised tasks, and the most widely used in particle physics. Chapter 4 is entirely dedicated to the jet tagging classification task.

- *Unsupervised learning* operates without labels, with the goal of learning structure in the input data. Generative models that learn to produce samples from p_{data} without access to class labels are a prominent example and are the focus of much of Chapter 5.

- *Self-supervised learning* is a hybrid approach in which the supervisory signal is derived from the data itself rather than from external annotation. This is most commonly used to learn general-purpose representations rather than to solve a specific downstream task. In HEP, self-supervised pretraining underlies some recent foundation models for jet physics, which use techniques such as masked particle modeling to learn representations of jet constituents that transfer across tasks [86, 87].

- *Reinforcement learning* (RL) trains an agent to maximize a reward signal through interaction with an environment. The key feature of RL is that it does not require a differentiable training objective, though some methods construct a surrogate in practice. RL is responsible for landmark results in game playing [24] and is the basis for the post-training pipelines of modern large language models [38]. Direct applications of RL within HEP have so far been limited [88], though its relevance is growing alongside the agentic AI systems discussed in Chapter 7.

Machine learning is not a new idea in particle physics. Boosted decision trees (BDTs) have been in use in HEP since 2004, and were applied to the single top quark discovery at the Tevatron [89, 90].[2] What changed with the deep learning era was not the concept of learned classifiers but the scale of the models. Neural networks scale better than BDTs, and so could handle the more complicated tasks that have driven much of the interest in ML in recent years.

The success of machine learning in HEP rests on two structural advantages. The first is access to high-quality simulated data produced by the Monte Carlo event generators discussed in Section 2.1. These simulated datasets carry *truth level* information that can be used to derive effective training labels. This is a significant advantage over other application domains where manual labeling is a significant bottleneck. Label assignment in HEP has its own subtleties, which are discussed in the context of top tagging in Chapter 4, but effective labels can often be set using straightforward generator-level rules. The second advantage is that HEP data is generated by physical processes that are well understood and are known to respect specific symmetries. Lorentz invariance, permutation invariance of jet constituents, and translational equivariance in the (η, ϕ) plane are all properties that have been used to aid neural network optimization. These properties could equally well be learned from the data and the community is starting to explore this direction, but exploiting them was a major benefit to early deep learning applications in HEP.

[2] The history of BDTs in HEP is interesting. The original application of BDTs illustrated that they *outperformed* artificial neural networks. Following the deep learning revolution initiated by AlexNet, BDTs were largely supplanted by neural networks, which proved far more expressive on the high-dimensional data common in HEP. Quite recently, BDTs have seen something of a resurgence in recent years, driven by their remarkable training speed relative to neural networks, and their robustness against noisy or irrelevant features in the density ratio estimation tasks covered in Section 3.5 [91].

Given that deep neural networks are the dominant paradigm in machine learning at present, the remainder of this chapter provides an overview of the core network architectures and training methods used throughout this thesis.

3.2 Neural Networks and Deep Learning

3.3 Generative Models

3.4 Diffusion Models

3.5 Density Ratio Estimation and Reweighting

3.6 Uncertainty Quantification

Chapter 4

Jet Classification at the LHC

4.1 Boosted Jet Tagging: Motivation and Landscape

4.2 ATLAS Top Tagger: Training and Uncertainty Quantification

4.3 The Future of Jet Classification

Chapter 5

Unfolding

5.1 Unfolding as an Inverse Problem

5.2 AI-Based Unfolding: Unbinned and High-Dimensional

5.3 Variational Latent Diffusion Unfolding

5.4 Omnifold

Chapter 6

Z+Jets Cross Section Measurement

6.1 Measurement Overview

6.2 Event Selection and Simulation Samples

6.3 OmniFold Implementation

6.4 Results: Four Physics Use Cases

6.4.1 Use Case 1: [FILL IN Title]

6.4.2 Use Case 2: [FILL IN Title]

6.4.3 Use Case 3: [FILL IN Title]

6.4.4 Use Case 4: [FILL IN Title]

6.5 Discussion

Chapter 7

Conclusion

7.1 Machine Learning is Driving LHC Physics Progress

7.2 diHiggs: The Next Milestone

7.3 Agentic AI and the Transformation of Physics Research

7.4 Will AI Deliver the Next Particle Physics Discovery?

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Appendix A

Supplementary Material

Appendix content goes here.